

Malhar Gudekar

(217) -766-0681 — gudekar2@illinois.edu — LinkedIn — Portfolio — GitHub

EDUCATION

University of Illinois Urbana Champaign

August 2024 - May 2026

Master of Science, Information Management

GPA: 3.8/4

• Coursework: Data Warehousing & Business Intelligence, Information Management, Information Consulting, Methods of Data Science

University of Mumbai

June 2019 - May 2023

Bachelor of Engineering, Computer Science

GPA: 9.25/10

• Coursework: DBMS, Big Data Analysis, Data Warehousing and Mining, Project Management, NLP, Machine Learning, Social Media Analytics

SKILLS

• **Analytical:** Data Analysis, Exploratory Data Analysis, Statistical Analysis, KPI Reporting, Data Validation, Data Cleaning, Anomaly Detection, ETL, Predictive Analytics, Natural Language Processing, Generative AI, Machine Learning, Time Series Forecasting, ARIMA/SARIMA

• **Programming / Querying:** Python, SQL, Pandas, NumPy, PySpark, PostgreSQL, Neo4j

• **Visualization / Reporting:** Power BI, Tableau, DAX, Excel, VBA, Dashboard Development, Data Visualization

• **Data / Cloud / Tools:** AWS S3, EC2, Redshift, EKS, FastAPI, Docker, Git, Kafka, pgVector, GCP

WORK EXPERIENCE

University of Illinois Urbana Champaign

May 2025 - Present

Research Assistant - iSchool

Champaign, USA

- Owned end-to-end design and development of cross-platform system integrating IoT sensor data for 100+ users, enabling scalable data ingestion and predictive analytics
- Designed and optimized **PostgreSQL architecture** for high-volume, production-scale datasets, improving query performance by 38%
- Built robust, production-ready **Python** pipeline with retrieval and context summarization, improving multi-turn response accuracy by 43%
- Developed backend workflows for real-time, fault-tolerant processing of time-series sensor data, enabling reliable monitoring and predictive analytics at scale

Research Assistant - CHI AI in Infodemic Management

January 2025 - May 2025

- Designed and deployed highly scalable natural language processing pipelines in Python to analyze and classify high-volume text datasets, reducing latency by 41%
- Built distributed data ingestion pipelines using **Kafka** and **Neo4j** Streams for real-time, high-throughput processing and pattern detection
- Developed production-grade **APIs** and deployment workflows to improve system reliability and reproducibility
- Developed technical documentation and reports to communicate AI system findings and workflows to both technical and non-technical stakeholders

Business Intelligence Group

August 2025 - December 2025

Technical Consultant

Champaign, USA

- Built and deployed scalable Generative AI and RAG system using FastAPI, PostgreSQL, and vector embeddings for production AI-driven knowledge retrieval
- Designed efficient, fault-tolerant **ETL** pipelines to process and index large document datasets for low-latency retrieval
- Improved system performance by 64% through optimized indexing strategies and pipeline efficiency improvements
- Developed backend services enabling real-time, high-performance querying of unstructured data

Swift Mobil Software Solutions Provider

July 2023 - October 2023

Data Analyst

Mumbai, IN

- Built interactive **Power BI** dashboards for **logistics** and **mobility KPIs**, reducing reporting time by 40% and improving visibility into operational performance across transportation services
- Processed and optimized large-scale operational datasets using **Python** and **PySpark** to identify failure, trends, and performance issues
- Developed **SQL** and **graph queries** to map dependencies, analyze operations, detect data quality issues, speeding root-cause analysis by 26%

Pscope Technologies Pvt. Ltd.

January 2023 - June 2023

Data Analyst

Mumbai, IN

- Developed and maintained 20+ **Power BI** dashboards across Sales, Finance, and Operations for enterprise clients, applying **SQL**, **DAX**, and **statistical analysis** to surface KPI trends and support cross-functional business decision-making
- Automated recurring **Excel**-based reporting workflows using **VBA**, reducing report preparation time by 42% and improving data consistency across weekly reporting cycles

Stu/Dio at Illinois

August 2025 - Present

Project Manager

Champaign, USA

- Led cross-functional team of 6 to deliver product updates for DeepCover, game, using user feedback and performance insights to guide data-driven feature decisions
- Tracked **sprint** progress, delivery metrics, and **QA issues** through **JIRA**-based workflows, improving cross-team visibility

PROJECTS

FMCG Sales Demand Forecasting

March 2025

[View Project](#)

- Analyzed time-series demand data across categories and locations to identify trends, seasonal patterns, and key variables driving consumption forecasting decisions
- Built and evaluated time-series forecasting models including ARIMA, SARIMA, and regression to predict demand patterns and improve load forecast accuracy using RMSE and MAPE
- Derived actionable insights on promotions, pricing, and stock behavior, enabling better inventory planning and reducing stockouts